Achieving Data Integrity and Confidentiality Using Image Steganography and Hashing Techniques

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***Abstract*—Most existing steganography algorithms keen on achieving data confidentiality only by embedding the data into a cover-media. This research paper introduced a new steganography technique that achieves both data confidentiality and integrity. Data confidentiality is achieved by embedding the data bits in a secret manner into stego-image. Integrity is achieved using SHA 256 hashing algorithm to hash the decoding and encoding variables. The proposed model performed a high PSNR values for using a dataset of different image sizes with an average PSNR of 82.933%.**

Keywords— Steganography, Data Confidentiality, Data Integrity, PSNR, SHA 256, Data Tampering.

I. INTRODUCTION

Image steganography is one of premier secure data hiding techniques. The role of steganography is to hide sensitive data into a cover image. This will protect the data from being captured by any unauthorized party. Steganography helps maintain data confidentiality, data integrity, data authentication, and data privacy. Steganography techniques vary depending on the algorithm used. Steganography can be combined with symmetric algorithms or asymmetric cryptography techniques. The advantage of a specific technique lies in the technique ability to achieve the information security fundamentals [1]. In this paper, a new steganography technique that combines a new approach for Least Significant Bit (LSB) method with a robust hash algorithm was introduced. The idea behind this approach is to embed any text into a cover image based on an offset flag, a shared key, and the robust hash function – SHA256 to achieve data confidentiality and integrity. Fig.1 illustrates the general process of steganography algorithm.

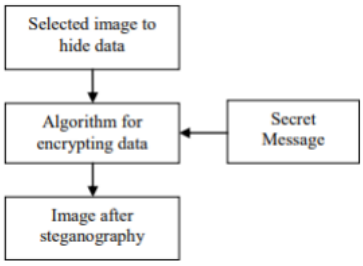

Fig.1. Steganography Workflow

A. Least Significant Bit Algorithm

All computer data is represented using binary, and grouped together in bytes. The LSB represents the right most bit of an 8 bits array and is associated with the lowest weight, while the Most Significant Bit (MSB) is the left most bit and is associated with the highest weight. The technique proposed in this paper is based on LSB, due to the minimal effect that LSB has on the original image. Typically, LSB encoding in steganography is performed by altering the LSB of the cover image to become similar to the value of the most significant bit of the plaintext.

B. Hash Function

Cryptographic hash function is a one way function that takes as input a variable length plaintext and generates a fixed size hash value. The hash function is accounted as a robust cryptography technique as it is infeasible to compute the plaintext. The hash function ensures that the sent plaintext is untampered by comparing the sent hash value with the decoded hash value. Secure Hash Algorithm (SHA) is one of the most commonly used hashing techniques. Many versions of the SHA algorithm have been introduced; the most popular family of hash function is the SHA-2 which was adopted in this research paper. SHA-2 consists of six hash functions with hash values: 224, 256, 384, or 512 bits. SHA-512 is separated into two sub-families; SHA-512/224 and SHA-512/256 [2].

The rest of this research paper is organized as follows: section 2 reviews related data encryption works that exploit steganography and hashing function in data encryption. The proposed technique is described in section 3. The performance measures of the designed model are presented in section 4. The results are detailed in section 5 and discussed in section 6 under security and performance analysis. Section 7 concludes the paper and provides areas for future research.

II. RELATED WORK

The vast majority of published researches focus on securing data transmission using different cryptography techniques. Steganography is an art that researchers have adopted for its robustness in encrypting sensitive data and achieving data confidentiality and integrity.

Jose et al. [3] adopted a new model to embed sensitive data into a cover-image by propagating the plaintext bits over the cover-image using hash salt technique with a password provided by the user. The adopted model increased the difficulties of brute-force attack as the salt hash will result into 2^{256} combinations of possible salted passwords. The authors also increased the model security by using Advanced

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Encryption Standard (AES) algorithm to encrypt the plaintext before embedding it into a cover-image.

Gupta et al. [4] proposed a hybrid approach by combining steganography with the AES algorithm and then used a hash function to increase the security of the model. The approach the authors proposed start by encrypting the plaintext using AES algorithm. The encrypted data then is stored in a hashed pixel location of the cover-image to generate a stego-image. The results of the proposed model achieved an accurate Mean Squared Error (MSE) values when evaluated on different image types such as, .tiff, .png, .jpg, .bmp, and .gif.

Chaudhary et al. [5] proposed an effective steganography technique that uses RGB images. The idea behind their work is to indicate the pixel value using the most significant bit of RGB channels instead of utilizing the entire channel. The algorithm works as follows: the LSB channels that are used for hidden data depend on the MSB sequence. For example, if the sequence of MSB is 101 then the data hidden sequence is GRB. Hash function was adopted in this model by applying the logical operator XOR between the cover-image LSB bits and the stego-image to indicate which pixels have changed.

Madhuravani et al. [6] presented an authenticated steganography scheme that uses a dynamic hashing algorithm. Firstly, the texture data is embedded into a cover-image that is then encoded using a stego-key. When the second party receives the stego-image, the stego-image is extracted to generate both the plaintext and the image-size. The plaintext after the extracting process is applied on a dynamic hash function using either MD5 or SHA functions to generate a digested text. The digested text is then embedded into a cover-image to generate the stego-image at receiver side. Once the stego-image is sent and received by the sender party, the receiver will extract the message and compares the received hash with the hash of extracted message. This new approach improved the security of steganography technique by securing the communication channel.

Riasat, et al. [7] introduced a robust hash-based steganography model. The designed model has a strenuous capability to hide image and data without losing the image quality. The data scattering depends on a random number that is generated by using a hash function, where the hash function uses both elements; the hash-key and the image chunks number. The image chunks are separated into three fields, where the ASCII values will distributed on these chunks sequentially.

Charan et al. [8] proposed an efficient secured steganography technique using multi-level encryption algorithms. The adopted model is based on two levels of encryption, the Chaos encryption and the Ceaser encryption techniques. At the beginning, the data is encrypted using Ceaser cipher. After the encryption, LSB algorithm is used and applied on the RGB image. Thus, the data distribution is done sequentially, where the first three bits are replaced in the three LSB bits of Red byte, the second three bits are replaced in the three LSB bits of Green, and the last two bits are replaced in the two LSB bits of Blue. The data after applying LSB are scattered into a Chaos cover-image that is divided into a two dimensional array and then the logical operator XOR is applied between the LSB bits and the array indexes.

Indrayani et al. [9] adopted a new mp3 audio steganography technique by combining steganography with AES algorithm and MD5 hash function. The designed model is divided into four core levels: encrypting the data using AES algorithm, where the key that is used to encrypt the data is digested using MD5 hash function. The encrypted data then are embedded into a cover-image which presents the encoding process. Once stego-image is received by the second party of communication, the image is extracted and the cipher text is decrypted using the same hashed-key. The authors achieved a high secured model against several active attacks types.

Saini et al. [10] proposed a hybrid approach in image security. The image is encrypted using a modified AES algorithm and then embedded into a cover-image to generate a stego-object. A new version of AES algorithm is presented "MAES"; where a new shift row transformation is presented. This transformation is done as follows: if the bit value that is located in the first row and first column of the initial matrix is even then no shifting is applied. Moreover, the other three rows will be shifted with an offset value equal to the offset value of the common AES row shifting transformation. The designed model achieves a high PSNR rate of all size images comparing to traditional AES algorithm.

The previous approaches focused on achieving a high secured steganography model using various algorithms of encryption and encoding. In this paper, a new steganography algorithm is presented to achieve confidentiality and integrity in a high performance secured model.

III. Proposed Model

This research paper introduces a hybrid steganography scheme that combines a new steganography algorithm approach with Hash function. The process of steganography is generally divided into two stages of encoding and decoding. In this research, the proposed model has four main stages: new image addressing, text size hashing, encoding, and decoding.

A. New Image Addressing and Confusion Concept

Image addressing process starts by selecting a conditional image size that will be used for embedding the plaintext. The reason behind using a conditional image size is that the following formula must apply.

$$\text{Image Size (IS)} = \text{Selected Image Size} - 512 \text{ pixels} \quad (1)$$

The intuition behind deducting 512 pixels is to store the needed variables, in which the 512 pixels are divided into two halves; the first half is reserved for the permutated encoded message size using the permutation formula as follows.

$$L = \begin{cases} i + IS, & i \bmod 2 = 0 \\ IS + 256 - i, & i \bmod 2 = 1 \end{cases} \quad (2)$$

Where L is Message Size Bits' Pixel Location Address.

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The second half is reserved for the hashed message size that will be discussed later. The permutation formula is based on the bit sequence state; if it is odd then it is stored starting from the top of the encoded cover-image as shown in equation (3), and, if the bit sequence is even then it is stored at the bottom of the encoded cover-image as shown in equation (4).

$$X = (I + K + \sqrt{MS + K}) \bmod IS \quad (3)$$

$$X = ((MS + K + \sqrt{MS + K}) - I) \bmod IS \quad (4)$$

Here *X* is the pixel location where to embed text bit in the stego-image (encoded cover-image), *I* represents the text bit sequence, *K* is the shared key between the parties of communication, *IS* represents the image size, and *MS* is the message size.

B. Text Size Hashing

First, the plain text was extracted into an array of bits that will be scattered over the encoded cover-image. The aim of adopting text size hashing is to increase the difficulty for any intruder to alter the text. Even if the texture is altered; the hash text size value indicates if any intentional alteration occurred. The text size was hashed using SHA-256 where the resulted hash value was applied on the logical operator XOR with the shared key *K* and then stored in the second half of 512 pixels of the image size as shown in the following equation.

$$H = (\text{Hashed Message Size}) \text{ XOR } K \quad (5)$$

C. Encoding

The encoding process in this research paper is divided into two sub-categories; the first encoding process was to calculate the address where the text bit will be embedded. Next, the encoding process was represented by embedding the text bits in the encoded cover-image to produce a stego-image, after distributing the hashed message size into the encoded cover-image. Fig. 2 illustrates the encoding process.

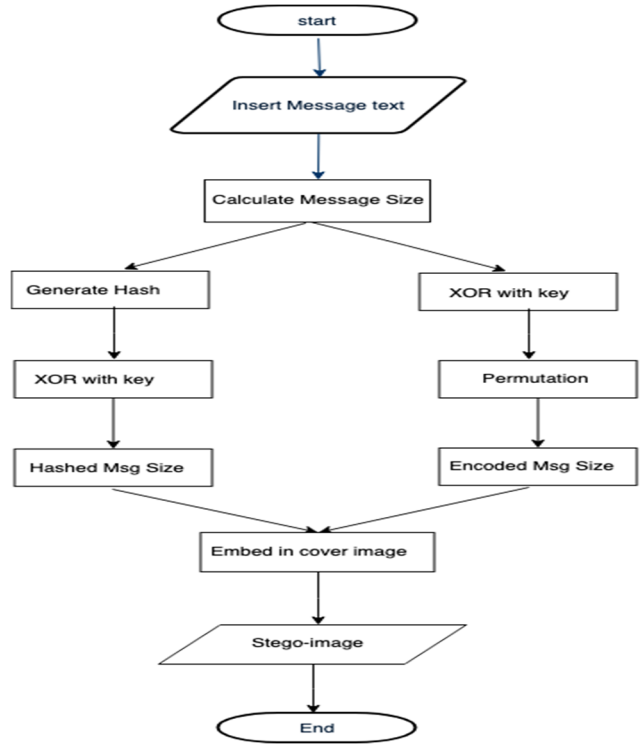

Fig.2. Encoding Process

D. Decoding

The decoding process is also divided into two sub-categories: once the stego-image is received by the second party of communication, the receiver retrieves the values of both halves that were located in the last 512 pixels; first of all, a retrieval of the encoded message size was performed using equations 3 and 4, then the receiver calculates the hashed message size value and apply the XOR logical operator with the shared key to ensure the equality between two values that ensures no data alteration occurred during the transmission.

The second sub-category process is how to extract the text message from the stego-image. The extracted characters are then stored in an array which is later converted to the original text (plaintext). Fig. 3 illustrates the decoding process.

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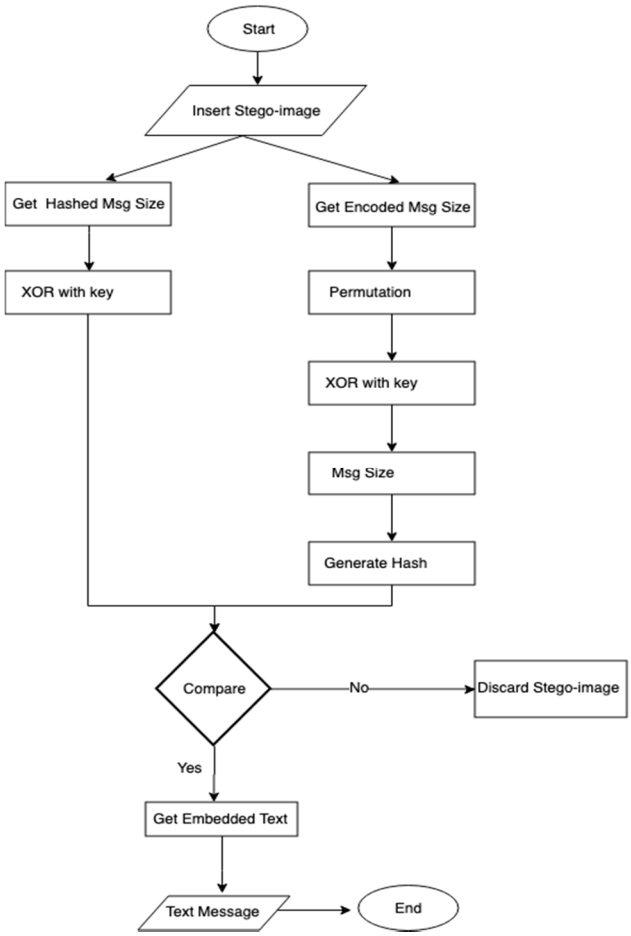

Fig.3. Decoding Process

IV. PERFORMANCE MEASURES

The proposed model performance depends on the efficiency rate of the overall implementation through the four stages. To ensure the efficiency of the proposed model the following rules are taken into consideration:

- The data must remain fixed after embedding it into the cover-image.
- The visual resolution of the image must remain high with no visual noise.
- The stego-image quality must be near to the original cover-image quality.
- The consumed time in encoding process is unremarkable.

Peak Signal to Noise Ratio (PSNR) is a well-known tool widely used to measure the visual quality of processed images. In this research, it is used to evaluate the stego-image quality and to compare it with the original cover-image quality. PSNR can be calculated as shown in eq 6; where a higher the PSNR dB indicates a better processed image quality

$$PSNR = 10 \log \frac{MAX^2}{MSE} \quad (6)$$

Where *MSE* is the mean square error between the processed and original cover image, Max^2 is the maximum intensity value used to represent each pixel in the image. If the obtained PSNR value is more than 30db then the processed image quality is unremarkably changed compared to the original. However, if the obtained PSNR value is less than 29db, this will result in a visual degradation in image quality. Table 1 illustrates a comparison in PSNR values between the proposed model and MAES algorithm [10]. The authors in MAES algorithm achieved a high PSNR values as well by combining steganography technique with an enhanced AES algorithm. The simulation results of the MAES were accurate enough to compare it the proposed model experimental results.

TABLE I. PSNR VALUES OF COVER-IMAGE VS STEGO-IMAGE

Algorithm	Image Size	PSNR (dB)
MAES [10]	128x128	59.1601
	256x256	59.9814
	512x512	59.4681
Proposed Model	128x128	78.6012
	256x256	82.1002
	512x512	87.4905

The overall performance of the proposed model is evaluated based on the time consumed in the encoding process. Attention is always given to ensure that the encoding process is completed in an acceptable time frame. Table 2 shows the time consumption of encoding characters into different image sizes.

TABLE II. PSNR VALUES OF COVER-IMAGE VS STEGO-IMAGE

Character Size	Image Size	Time (ms)
500 Character	128x128	42
	256x256	166
	512x512	532
1000 Character	128x128	241
	256x256	325
	512x512	632
1500 Character	128x128	513
	256x256	756
	512x512	852

Histogram analysis is well known used technique to measure changes between original and processed images. Histogram analysis calculates the levels of intensity values available in a given image. Therefore, it was used in this research. It resulted in minor differences between the original cover-image and the stego-image. Fig.4 illustrates the histogram analysis of three images that are used in the proposed model, which indicates that the stego-image can pass the histogram test.

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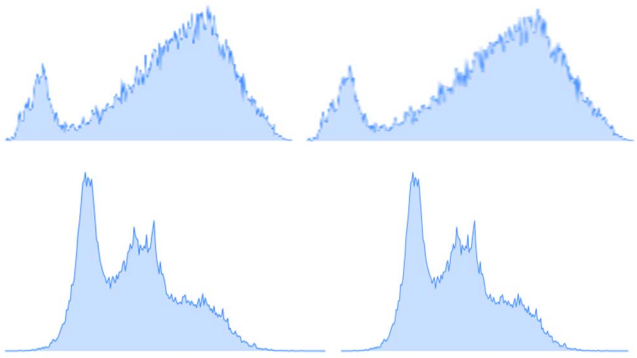

Fig. 4. (a) Cover-Image Histogram Vs. (B) Stego-Image Histogram

The histogram analysis shows a slight difference between cover-images that are selected for text embedding and the resulting stego-images. This indicates that the user can not recognize any difference between both images resolutions.

V. Expermental Results and Analysis

The proposed model was measured using Intel® Core™ i7-4580HQ 64 bits system with 8GB RAM running on windows 8.1.

Different image sizes (128x128, 256x256, and 512x512) were selected for the encoding process as illustrated in Fig.5 respectively.

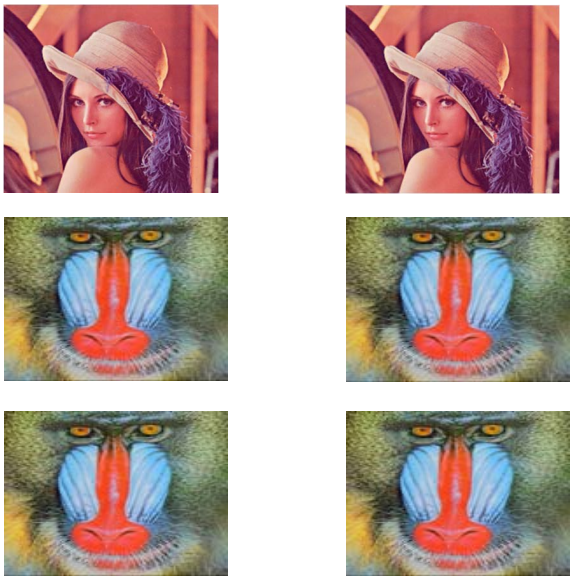

Fig. 5. (a) Cover-Image VS (B) Stego-Image

It was found that the resolution of both images remained the same. The results ensure that the embedded text did not have an effect on the resulting visual image quality.

The aim behind adopting the proposed work is to come up with a high performance and secured steganography scheme. The experimental results concluded two main analysis as follows.

A. Security Analysis

The proposed model achieved two core concepts in information security; confidentiality and integrity. Confidentiality was achieved by embedding the text bits into the LSB pixels of the cover-image using formulas that depends on a secret key. The text size was hashed SHA-256 algorithm and then applied it to the logical operator XOR that produced a hashed message size. This process achieves the concept of integrity. Using robust hash function increased the difficulties of brute-force attack; where the hashed message size is hard to be calculated due to the XOR operation with the secret shared key.

B. Performance Analysis

The proposed model fulfilled a high performance through a set of various metrics. The PSNR results that were discussed in table 1 had shown that the proposed model was better than the MAES algorithm. Moreover, the encoding and decoding processes were executed over Csharp programing language, concluding that the time that was consumed by the overall model is less than the consumed time while running the implementation over Matlab or other machine learning. At the end, the performance of the proposed model depends on the selected image size as it found that the stego-image size is approximately equal to the cover-image size.

VI. Conclusion and Future Work

The huge data transfer over public networks leads most of information security engineers to adopt many methods that allow them to transfer sensitive data over a secure fashion environment. Steganography is one of the most commonly used techniques, due to the ease of use and the high data security that can be provided. In this paper the combined approach for image security has been presented. The scattering and embedding processes of texture data were achieved over a set of equations. The proposed model increased the difficulties for any intruder to alter the embedded sensitive data which proved over histogram analysis that shows a slight difference between cover-images that are selected for text embedding and the resulting stego-images, as the concepts of confusion, permutation, and hashing were adopted in the discussed model. For future work, the selected image type should be expanding to include different types of images such as, .tiff, .bmp, and .gif.

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